TEST-I ACM Dated 4.9.2023 MM.15

Attempt any **three** questions. Qno 1 is compulsory. Attempt any **five** questions from Qno.1

Qn1(a) What are the advantage of p.u system.

(b) Why zero voltage regulation is only possible at leading power factor.

(c) What is the condition for maximum efficiency of a transformer.

(d) Low power factor is used for OC test and High Power Factor Wattmeter is used for SC test.why?

(e) Why polarity test is essential condition for parallel operation of transformer?

(f) Explain advantage of Sumpener test

(g) Function of circulation current

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Qn2. Draw the phasor diagram of a single-phase transformer at lagging and leading power factor

Qn3. Prove that maximum voltage regulation is (Z_{e2})p.u

Qn4. The no-load current of a transformer is 5A at 0.25 p.f when supplied at 230V, 50 Hz. The number of turns on primary winding is 200. Calculate:

(i) Maximum value of flux in the core

(ii) Core Loss

(iii) Magnetizing Current